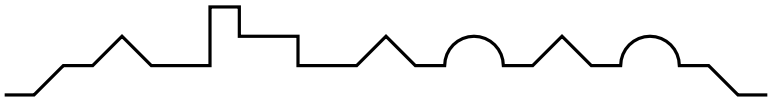
Math Kangaroo Practice

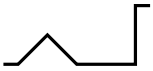
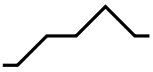
Grade 3-4 · 24 questions · harder · seed 47999

Part A: Each question is worth 3 points

1. This is the outline of a town seen from the river at sunset.

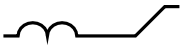
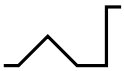
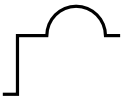


Which piece **cannot** be part of the outline?

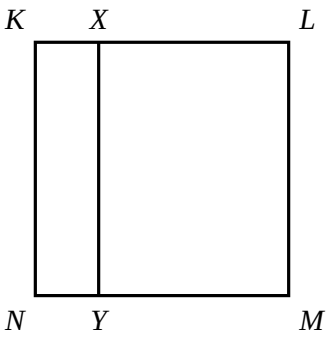
A.  ***B.*** ***C.*** ***D.*** ***E.*** 

2. Far away we see the skyline of a castle.

Which of the pieces **cannot** belong to the skyline?

A.  ***B.***  ***C.*** ***D.*** ***E.*** 

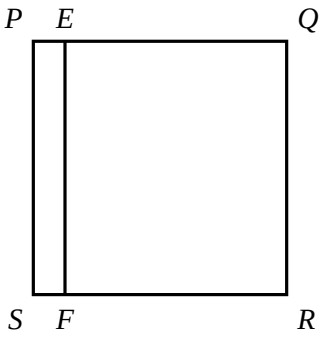
3. *KLMN* is a square. Its side is 12 cm. The line *XY* cuts it into two rectangles, and *KX* = 3 cm.



How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *XLMY*?

A. 42 cm ***B.*** 24 cm ***C.*** 12 cm ***D.*** 6 cm ***E.*** 3 cm

4. *PQRS* is a square. Its side is 16 cm. The line *EF* cuts it into two rectangles, and *PE* = 2 cm.

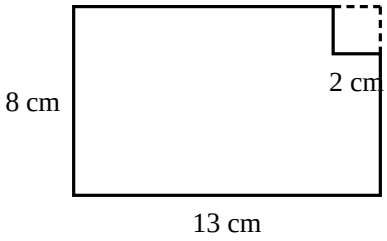


How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *EQRF*?

A. 28 cm ***B.*** 4 cm ***C.*** 3 cm ***D.*** 2 cm ***E.*** 1 cm

Part B: Each question is worth 4 points

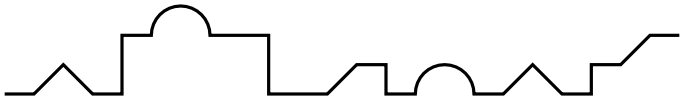
5. A rectangle is 13 cm long and 8 cm wide. A square with side 2 cm is cut off one corner.



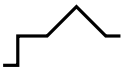
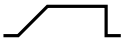
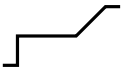
What is the perimeter of the remaining shape?

A. 50 cm ***B.*** 46 cm ***C.*** 42 cm ***D.*** 40 cm ***E.*** 21 cm

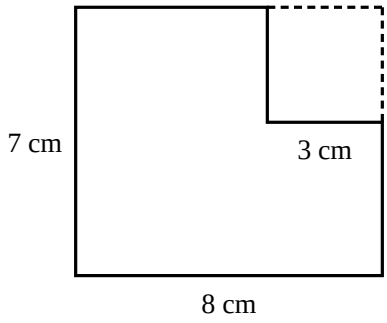
6. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A. ***B.*** ***C.***  ***D.***  ***E.*** 

7. A rectangle is 8 cm long and 7 cm wide. A square with side 3 cm is cut off one corner.

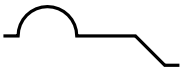
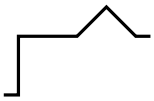


What is the perimeter of the remaining shape?

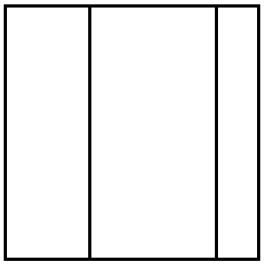
A. 42 cm ***B.*** 36 cm ***C.*** 34 cm ***D.*** 31 cm ***E.*** 30 cm

8. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A.  ***B.***  ***C.*** ***D.*** ***E.***

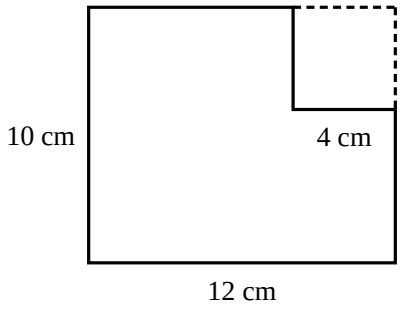
9. A square garden is surrounded by a fence 32 m long. The gardener divides it into 3 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 96 m ***B.*** 77 m ***C.*** 74 m ***D.*** 64 m ***E.*** 32 m

10. A rectangle is 12 cm long and 10 cm wide. A square with side 4 cm is cut off one corner.

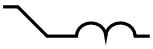
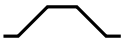
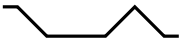


What is the perimeter of the remaining shape?

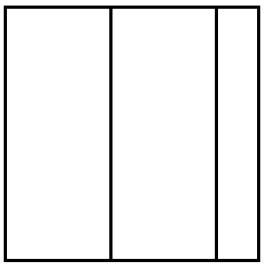
A. 44 cm ***B.*** 47 cm ***C.*** 48 cm ***D.*** 52 cm ***E.*** 60 cm

11. This is the outline of a town seen from the river at sunset.

Which piece **can** be part of the outline?

A. ***B.*** ***C.***  ***D.***  ***E.*** 

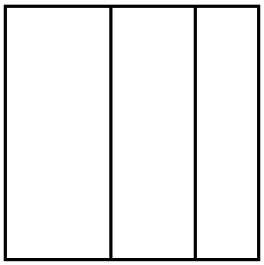
12. A square garden is surrounded by a fence 68 m long. The gardener divides it into 3 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 204 m ***B.*** 136 m ***C.*** 119 m ***D.*** 102 m ***E.*** 68 m

13. Grandma bakes a square cake. A ribbon 52 cm long fits exactly around it. She cuts the cake into 3 rectangular pieces as shown and puts a ribbon around every piece.

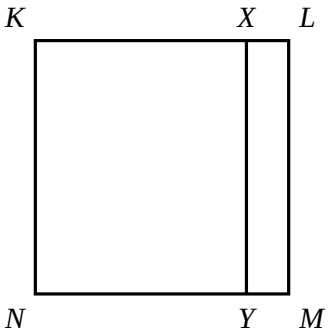


How many centimetres of ribbon does she need for all the pieces together?

A. 104 cm ***B.*** 91 cm ***C.*** 89 cm ***D.*** 78 cm ***E.*** 52 cm

Part C: Each question is worth 5 points

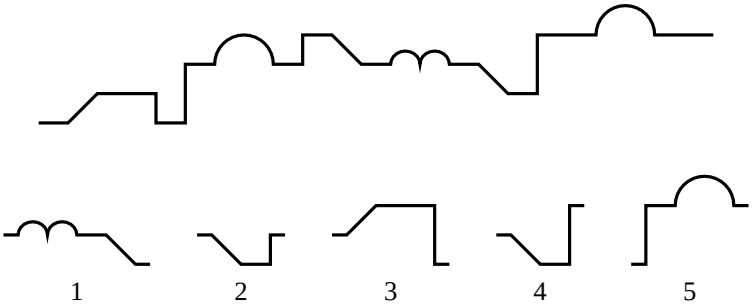
14. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 44 cm and the perimeter of *XLMY* is 28 cm.



What is the perimeter of the square *KLMN*?

A. 48 cm ***B.*** 36 cm ***C.*** 18 cm ***D.*** 16 cm ***E.*** 12 cm

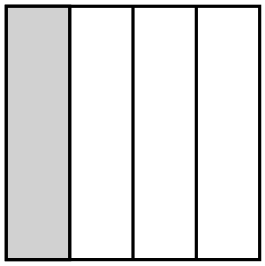
15. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 4 ***E.*** 5

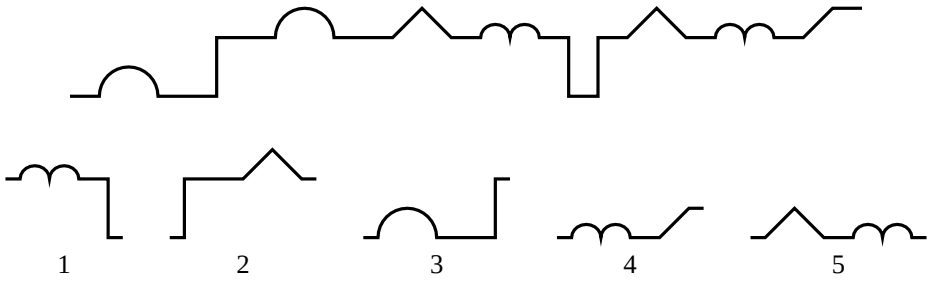
16. Mia cuts a square chocolate bar into 4 equal strips as shown. The perimeter of each strip is 90 cm.



What is the perimeter of the square?

A. 360 cm ***B.*** 144 cm ***C.*** 108 cm ***D.*** 72 cm ***E.*** 36 cm

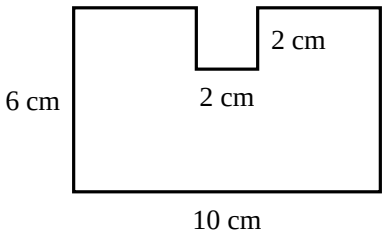
17. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

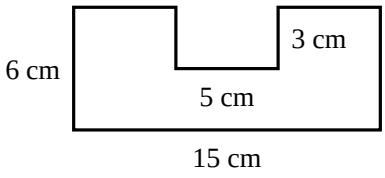
18. A 10 cm by 6 cm rectangle of cardboard has a rectangular notch 2 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 40 cm ***B.*** 38 cm ***C.*** 37 cm ***D.*** 36 cm ***E.*** 30 cm

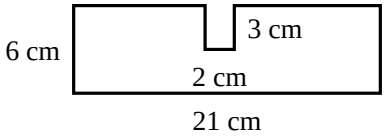
19. A 15 cm by 6 cm rectangle of cardboard has a rectangular notch 5 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 37 cm ***B.*** 43 cm ***C.*** 45 cm ***D.*** 48 cm ***E.*** 52 cm

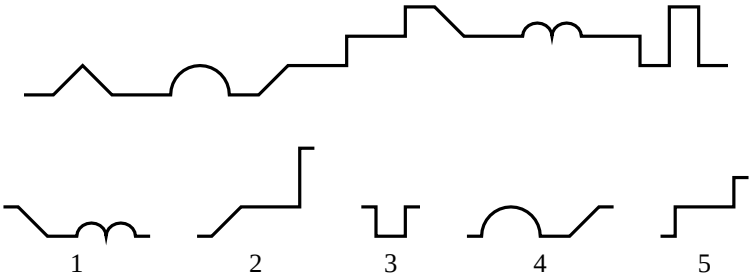
20. A 21 cm by 6 cm rectangle of cardboard has a rectangular notch 2 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 60 cm ***B.*** 62 cm ***C.*** 64 cm ***D.*** 74 cm ***E.*** 75 cm

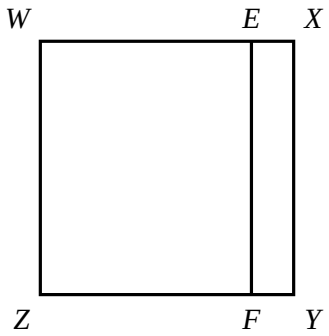
21. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

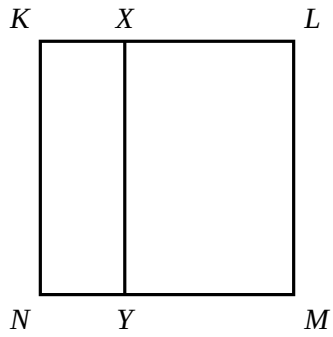
22. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 22 cm and the perimeter of *EXYF* is 14 cm.



How long is the side of the square *WXYZ*?

A. 30 cm ***B.*** 12 cm ***C.*** 8 cm ***D.*** 6 cm ***E.*** 5 cm

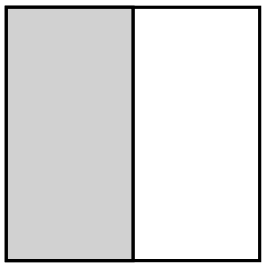
23. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 32 cm and the perimeter of *XLMY* is 40 cm.



What is the perimeter of the square *KLMN*?

A. 8 cm ***B.*** 18 cm ***C.*** 24 cm ***D.*** 36 cm ***E.*** 48 cm

24. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 24 cm.



What is the perimeter of the square?

A. 32 cm ***B.*** 30 cm ***C.*** 26 cm ***D.*** 16 cm ***E.*** 8 cm

Answer Key

1.	**B**	**2.**	**A**	**3.**	**D**	**4.**	**B**	**5.**	**C**	**6.**	**A**
7.	**E**	**8.**	**D**	**9.**	**D**	**10.**	**A**	**11.**	**D**	**12.**	**B**
13.	**A**	**14.**	**A**	**15.**	**C**	**16.**	**B**	**17.**	**D**	**18.**	**D**
19.	**D**	**20.**	**A**	**21.**	**C**	**22.**	**D**	**23.**	**E**	**24.**	**A**

Solutions

1. B — All the other pieces can be found in the line, but this one cannot: it is a mirror image of a part of the line. [2002 Q7 · cannot_belong]

2. A — All the other pieces can be found in the line, but this one cannot: its flat part between the two features has the wrong length. [2002 Q7 · cannot_belong]

3. D (6 cm) — *XLMY* has sides 9 and 12 cm, perimeter 42 cm. Square: 48 cm. Difference 6 cm (= 2 × 3). [2002 Q6 · square_minus_strip]

4. B (4 cm) — *EQRF* has sides 14 and 16 cm, perimeter 60 cm. Square: 64 cm. Difference 4 cm (= 2 × 2). [2002 Q6 · square_minus_strip]

5. C (42 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (13 + 8) = 42 cm. [2002 Q6 · corner_cut]

6. A — Only this piece is found in the line. The others don't fit — B: it has the right shapes but in the wrong order; C: it has the right shapes but in the wrong order; D: its flat part between the two features has the wrong length; E: its flat part between the two features has the wrong length. [2002 Q7 · can_belong]

7. E (30 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (8 + 7) = 30 cm. [2002 Q6 · corner_cut]

8. D — Only this piece is found in the line. The others don't fit — A: its flat part between the two features has the wrong length; B: it has a pointed roof where the line has a round dome; C: its step is taller than in the line; E: it is a mirror image of a part of the line. [2002 Q7 · can_belong]

9. D (64 m) — Side = 32 ÷ 4 = 8 m. There are 2 cuts; each cut is 8 m long and is counted twice (once for each piece next to it): 32 + 2 × 2 × 8 = 64 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

10. A (44 cm) — The two new edges (4 cm and 4 cm) replace the two removed pieces of the outer edge, which were also 4 cm and 4 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 10) = 44 cm. [2002 Q6 · corner_cut]

11. D — Only this piece is found in the line. The others don't fit — A: its flat part between the two features has the wrong length; B: it has the right shapes but in the wrong order; C: its flat part between the two features has the wrong length; E: it has a pointed roof where the line has two small bumps. [2002 Q7 · can_belong]

12. B (136 m) — Side = 68 ÷ 4 = 17 m. There are 2 cuts; each cut is 17 m long and is counted twice (once for each piece next to it): 68 + 2 × 2 × 17 = 136 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

13. A (104 cm) — Side = 52 ÷ 4 = 13 cm. There are 2 cuts; each cut is 13 cm long and is counted twice (once for each piece next to it): 52 + 2 × 2 × 13 = 104 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

14. A (48 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 44 + 28 = 72 cm = 6 sides. Side = 72 ÷ 6 = 12 cm, perimeter 4 × 12 = 48 cm. [2002 Q6 · two_perimeters]

15. C (3) — Pieces 1, 4, 5 fit. Not: 2 (its step is shorter than in the line); 3 (its step is taller than in the line). [2002 Q7 · how_many_fit]

16. B (144 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 90 ÷ 10 = 9, so the side is 4 × 9 = 36 cm and the perimeter is 144 cm. [2002 Q6 · strips_to_square]

17. D (4) — Pieces 1, 3, 4, 5 fit. Not: 2 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

18. D (36 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (10 + 6) + 2 × 2 = 36 cm. [2002 Q6 · notch]

19. D (48 cm) — The bottom of the notch (5 cm) replaces the 5 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (15 + 6) + 2 × 3 = 48 cm. [2002 Q6 · notch]

20. A (60 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (21 + 6) + 2 × 3 = 60 cm. [2002 Q6 · notch]

21. C (2) — Pieces 4, 5 fit. Not: 1 (its flat part between the two features has the wrong length); 2 (its step is taller than in the line); 3 (its step is shorter than in the line). [2002 Q7 · how_many_fit]

22. D (6 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 22 + 14 = 36 cm = 6 sides. Side = 36 ÷ 6 = 6 cm. [2002 Q6 · two_perimeters]

23. E (48 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 32 + 40 = 72 cm = 6 sides. Side = 72 ÷ 6 = 12 cm, perimeter 4 × 12 = 48 cm. [2002 Q6 · two_perimeters]

24. A (32 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 24 ÷ 6 = 4, so the side is 2 × 4 = 8 cm and the perimeter is 32 cm. [2002 Q6 · strips_to_square]